

CHAPTER 9: BIOMOLECULES

Exhaustive Academic & Exam-Oriented Notes • Preserving 100% Textbook Sequence, Concepts, Equations, Tables, and Diagrams.

Source: NCERT Biology Textbook

Target: NEET / Board Exams / Comprehensive Study

Format: High-Density Landscape Grid

☰ QUICK NAVIGATION INDEX

🧠 Mind Map

9.1 Chemical Analysis

9.2 Metabolites

9.3 Biomacromolecules

9.4 Proteins

9.5 Polysaccharides

9.6 Nucleic Acids

9.7 Protein Structure

9.8 Enzymes

📖 Final Revision

📄 NCERT Exercises

SECTION 9.0



CHAPTER MIND MAP — BIOMOLECULES OVERVIEW

1. Chemical Composition Analysis

- **Grinding:** Tissue + Cl_3CCOOH
- **Acid-Soluble Pool:** Micromolecules (18-800 Da), Cytoplasmic composition
- **Acid-Insoluble Retentate:** Macromolecules (>10,000 Da) + Lipids
- **Ash Analysis:** Destructive technique for inorganic elements (Ca^{2+} , Mg^{2+} , PO_4^{3-})

2. Metabolites & Biomacromolecules

- **Primary Metabolites:** Direct physiological roles (Amino acids, Sugars)
- **Secondary Metabolites:** Plant/Microbial products (Alkaloids, Flavonoids, Rubber, Drugs)
- **Lipid Exception:** MW <800 Da but insoluble due to cell membrane vesicles
- **Cell Mass:** Water (70-90%), Protein (10-15%), Nucleic Acid (5-7%)

3. Polymers & Structures

- **Proteins:** Heteropolymers of 20 α -amino acids (Peptide bonds). Collagen & RuBisCO
- **Polysaccharides:** Homopolymers (Cellulose, Starch, Glycogen, Chitin) & Inulin (Fructose)
- **Nucleic Acids:** Polynucleotides (DNA/RNA) built of Nitrogen base + Sugar + Phosphate
- **Protein Hierarchy:** Primary $\rightarrow$ Secondary (α/β) $\rightarrow$ Tertiary (3D) $\rightarrow$ Quaternary (Hb)

4. Enzymes & Catalysis

- **Nature:** Biological catalysts (Proteins/Ribozymes) with Active Site crevices
- **Mechanism:** Lowers Activation Energy barrier ($E + S \rightleftharpoons ES \rightarrow EP \rightarrow E + P$)
- **Factors:** Temp & pH Optima, [S] saturation (V_{max} , K_m), Competitive Inhibition
- **Classification:** 6 Classes (Oxidoreductase, Transferase, Hydrolase, Lyase, Isomerase, Ligase)
- **Co-factors:** Apoenzyme + (Prosthetic group / Co-enzyme / Metal ion)

SECTION 9.1 HOW TO ANALYSE CHEMICAL COMPOSITION?

NCERT Pages 104-107

Chemical Analysis Protocol (Tissue Fractionation)

1. **Step 1:** Take living tissue (vegetable leaf or liver tissue).
2. **Step 2:** Grind in **Trichloroacetic Acid (Cl_3CCOOH)** using a mortar and pestle to obtain a thick slurry.
3. **Step 3:** Strain slurry through cheesecloth or cotton wool.
4. **Step 4:** Collect two distinct chemical fractions:

- **Filtrate (Acid-Soluble Pool):** Contains thousands of organic micromolecules (MW 18 to 800 Da). Represents cytoplasmic composition.
- **Retentate (Acid-Insoluble Fraction):** Contains biomacromolecules (>10,000 Da) such as proteins, nucleic acids, polysaccharides, and lipid membrane vesicles.

🔥 Inorganic "Ash" Analysis (Destructive Method):

Weigh fresh tissue (**Wet Weight**) → Dry tissue to evaporate water (**Dry Weight**) → Fully burn tissue to oxidize carbon compounds to gases (CO₂, water vapor) → Remaining matter is **ASH** containing inorganic elements (Ca²⁺, Mg²⁺, Na⁺, K⁺) and compounds (PO₄³⁻, SO₄²⁻, NaCl, CaCO₃).

TABLE 9.1: COMPARISON OF ELEMENTS IN NON-LIVING VS LIVING MATTER

Element	% Weight in Earth's Crust	% Weight in Human Body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

*Relative abundance of Carbon (C) and Hydrogen (H) is significantly higher in living organisms than in Earth's crust.

TABLE 9.2: INORGANIC CONSTITUENTS OF LIVING TISSUES

Component	Formula / Representation
Sodium	Na ⁺
Potassium	K ⁺
Calcium	Ca ²⁺
Magnesium	Mg ²⁺
Water	H ₂ O
Compounds	NaCl, CaCO ₃ , PO ₄ ³⁻ , SO ₄ ²⁻

★ EXAM KEY POINT

Inorganic compounds like sulphate (SO₄²⁻) and phosphate (PO₄³⁻) are present in the **acid-soluble fraction** during elemental analysis.

A. AMINO ACIDS (α-amino acids)

Substituted Methanes

Organic compounds containing an **amino group** (–NH₂) and a **carboxyl group** (–COOH) as substituents on the same carbon atom (the **α-carbon**).

Four Substituents on Valency Positions:

1. Hydrogen (-H) | 2. Carboxyl group (-COOH) | 3. Amino group (-NH₂) | 4. Variable group (-R)

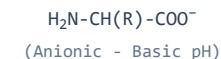
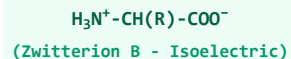
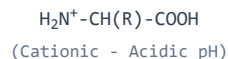
Types of Amino Acids based on R-Group & Properties:

- **Glycine:** R = -H (Simplest, Achiral)
- **Alanine:** R = -CH₃ (Methyl group)
- **Serine:** R = -CH₂-OH (Hydroxy methyl group)
- **Acidic:** Glutamic acid (Extra -COOH)
- **Basic:** Lysine (Extra -NH₂)
- **Neutral:** Valine
- **Aromatic:** Tyrosine, Phenylalanine, Tryptophan

⚡ Zwitterionic Form of Amino Acids

pH Dependent

Due to the ionizable nature of -NH₂ and -COOH groups, the structure of amino acids changes in solutions of different pH.



B. LIPIDS (Water Insoluble Biomolecules)

MW < 800 Da

Lipids are water-insoluble, small molecular weight compounds comprising fatty acids, glycerol esters, and phospholipids.

1. Fatty Acids (R-COOH):

R group contains 1 to 19 carbons. Examples:

- **Palmitic Acid:** 16 Carbons (including carboxyl carbon).
- **Arachidonic Acid:** 20 Carbons (including carboxyl carbon).
- **Saturated:** Without C=C double bonds.
- **Unsaturated:** With one or more C=C double bonds.

2. Glycerol & Glycerides:

Glycerol: Trihydroxy propane. Fatty acids esterified with glycerol yield **Monoglycerides, Diglycerides, and Triglycerides**.

Fats vs Oils: Based on melting point. Oils (e.g., *gingelly oil*) have lower melting point and remain liquid in winter.

3. Phospholipids & Complex Lipids:

Contains phosphorus and phosphorylated organic compound. Found in cell membranes. Key Example: **Lecithin**.

Neural tissues contain lipids with highly complex structures.

C. NITROGEN BASES, NUCLEOSIDES & NUCLEOTIDES

Heterocyclic Rings

Carbon compounds featuring heterocyclic nitrogen rings form the foundational components of nucleic acids (DNA & RNA).

Category	Chemical Constituents / Examples
Nitrogen Bases	Purines (Double Ring): Adenine (A), Guanine (G) Pyrimidines (Single Ring): Cytosine (C), Uracil (U), Thymine (T)
Nucleosides (Base + Sugar)	Adenosine, Guanosine, Thymidine, Uridine, Cytidine
Nucleotides (Base + Sugar + Phosphate)	Adenylic acid, Guanylic acid, Thymidylic acid, Uridylic acid, Cytidylic acid

🧠 MEMORY TRICK: Base vs Nucleoside vs Nucleotide

- Base = **Adenine** (Pure nitrogen ring)
- Base + Sugar = **Adenosine** (Nucleoside)
- Base + Sugar + Phosphate = **Adenylic Acid** (Nucleotide ending in '-ic acid')

SECTION 9.2 PRIMARY AND SECONDARY METABOLITES

NCERT Pages 108

Metabolite Categorization

● Primary Metabolites

Present in animal tissues. Have **identifiable physiological functions** and play known direct roles in normal metabolic processes (e.g., amino acids, sugars, nucleotides, lipids).

● Secondary Metabolites

Found in plant, fungal, and microbial cells. Thousands of compounds other than primary metabolites. Roles in host organism's normal physiology are not fully understood, but many are vital for **human welfare** and **ecological interactions**.

💡 **Significance of Secondary Metabolites:**

- **Human Welfare:** Rubber, drugs, spices, scents, pigments, antibiotics.
- **Ecological Roles:** Defense mechanisms, attractants for pollinators, allelopathy.

TABLE 9.3: COMPREHENSIVE LIST OF SECONDARY METABOLITES (NCERT MANDATORY)

HIGH YIELD NEET TABLE

Category	Examples & Representative Compounds
Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoids	Monoterpenes, Diterpenes, etc.
Essential Oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastine, Curcumin, etc.
Polymeric Substances	Rubber, Gums, Cellulose

💡 **MEMORY TRICKS FOR HIGH-YIELD SECONDARY METABOLITES**

Toxins (AR): Abrin & Ricin

Lectins (C): Concanavalin A

Drugs (VC): Vinblastine & Curcumin

Biomolecules Classification by Molecular Weight

Biomolecules / Biomicromolecules:
Found in the acid-soluble pool. Molecular weight ranges from **18 to ~800 Daltons (Da)**.

Biomacromolecules:
Found in the acid-insoluble fraction. Molecular weight in the range of **10,000 Daltons and above**.
Four major classes: **Proteins, Nucleic Acids, Polysaccharides, and Lipids (Exception)**.

⚠ THE LIPID PARADOX (NEET FAVORITE) Why Lipids end up in Retentate?
Lipids have molecular weights **not exceeding 800 Da** (hence chemically small micromolecules). However, they are arranged into cell membranes and organelle structures. Upon grinding tissue, cell membranes disrupt into **water-insoluble vesicles** that separate with the acid-insoluble fraction. Thus, lipids are strictly **NOT biomacromolecules**, but are recovered in the macromolecular retentate!

TABLE 9.4: AVERAGE COMPOSITION OF CELLS (% OF TOTAL CELLULAR MASS)

Cellular Component	% of Total Cellular Mass	Macromolecule / Micromolecule Status
Water (H ₂ O)	70–90% (Most Abundant Chemical)	Inorganic Solvent / Medium
Proteins	10–15% (Most Abundant Organic Macromolecule)	True Biomacromolecule
Nucleic Acids	5–7%	True Biomacromolecule
Carbohydrates	3%	True Biomacromolecule (Polysaccharides)
Lipids	2%	Acid-insoluble Micromolecule
Ions	1%	Inorganic Constituents

Summary of Cellular Pools:

- **Acid-soluble pool:** Represents roughly the *cytoplasmic composition*.
- **Acid-insoluble pool:** Represents macromolecules from cytoplasm and organelles.
- Together, both fractions account for the *entire chemical composition* of living tissues.

SECTION 9.4 PROTEINS

NCERT Pages 109-110

Polymeric Nature of Proteins

Polypeptides: Proteins are linear chains of amino acids linked together by **peptide bonds**.

Heteropolymer: Protein is ALWAYS a **heteropolymer** and NOT a homopolymer because it consists of 20 different types of amino acid monomers (e.g., alanine, cysteine, proline, tryptophan, lysine).

Essential vs Non-Essential Amino Acids:

- **Essential:** Cannot be synthesized by our body; must be supplied through dietary food.
- **Non-essential:** Can be synthesized by the body.

🏆 BIOSPHERE ABUNDANCE RECORDS

- **Collagen:** Most abundant protein in the **animal world**.
- **RuBisCO:** (Ribulose bisphosphate Carboxylase-Oxygenase) Most abundant protein in the **whole biosphere**.

TABLE 9.5: REPRESENTATIVE PROTEINS AND THEIR BIOLOGICAL FUNCTIONS

Protein	Specific Biological Function
Collagen	Intercellular ground substance in connective tissues.
Trypsin	Digestive proteolytic enzyme.
Insulin	Peptide hormone regulating glucose homeostasis.

Protein	Specific Biological Function
Antibody	Protective immunoglobulin that fights infectious agents/pathogens.
Receptor	Sensory reception (smell, taste, hormone recognition, etc.).
GLUT-4	Glucose transporter enabling glucose transport across cell membranes into cells.

Peptide Bond Formation (–CO–NH–):

Formed when the carboxyl group (–COOH) of one amino acid reacts with the amino group (–NH₂) of the next amino acid with the elimination of a water molecule (H₂O dehydration reaction).

SECTION 9.5 POLYSACCHARIDES

NCERT Pages 110–111

Types & Polymer Structure

- **Cellulose:** Polymeric polysaccharide of *glucose* monomers. It is a **homopolymer**. Major component of plant cell walls, plant pulp paper, and cotton fibres.
- **Starch:** Homopolymer of glucose; energy storehouse in plant tissues. Forms **helical secondary structures**.
- **Glycogen:** Highly branched homopolymer of glucose; energy storehouse in animal tissues.
- **Inulin:** Polymer of **fructose** monomers.
- **Chitin:** Complex polysaccharide homopolymer forming exoskeletons of arthropods & fungal cell walls. Monomers are amino-sugars (*N*-acetyl glucosamine).

Glycogen Chain Ends & Helix Test

Glycogen Polysaccharide Directionality

Left End:
NON-REDUCING END

— Glucose Chain —▶

Right End:
REDUCING END

Starch vs Cellulose Iodine Test

Starch forms **secondary helical structures** that can trap/hold I_2 (Iodine) molecules inside the helices, giving a **blue color**. Cellulose does *not* contain complex helices and therefore **cannot hold I_2** (gives no blue color).

Complex Polysaccharides & Derivatives

Naturally occurring polysaccharides can contain chemically modified sugars as building blocks:

- **Amino-sugars:** Glucosamine, Galactosamine.
- **Modified Sugars:** *N*-acetyl glucosamine (NAG), *N*-acetyl galactosamine.
- **Exoskeletons:** Arthropod exoskeletons consist of **Chitin** (mostly homopolymer of NAG).

 **Bond Type:** Monosaccharides are linked together by **Glycosidic Bonds** formed by dehydration (loss of H_2O).

SECTION 9.5 POLYSACCHARIDES

NCERT Pages 110-111

Types & Polymer Structure

- **Cellulose:** Polymeric polysaccharide of *glucose* monomers. It is a **homopolymer**. Major component of plant cell walls, plant pulp paper, and cotton fibres.
- **Starch:** Homopolymer of glucose; energy storehouse in plant tissues. Forms **helical secondary structures**.
- **Glycogen:** Highly branched homopolymer of glucose; energy storehouse in animal tissues.
- **Inulin:** Polymer of **fructose** monomers.
- **Chitin:** Complex polysaccharide homopolymer forming exoskeletons of arthropods & fungal cell walls. Monomers are amino-sugars (*N*-acetyl glucosamine).

Glycogen Chain Ends & Helix Test

Glycogen Polysaccharide Directionality

Left End:
NON-REDUCING END

— Glucose Chain —▶

Right End:
REDUCING END

🔬 Starch vs Cellulose Iodine Test

Starch forms **secondary helical structures** that can trap/hold I_2 (Iodine) molecules inside the helices, giving a **blue color**.
Cellulose does *not* contain complex helices and therefore **cannot hold** I_2 (gives no blue color).

Complex Polysaccharides & Derivatives

Naturally occurring polysaccharides can contain chemically modified sugars as building blocks:

- **Amino-sugars:** Glucosamine, Galactosamine.
- **Modified Sugars:** *N*-acetyl glucosamine (NAG), *N*-acetyl galactosamine.
- **Exoskeletons:** Arthropod exoskeletons consist of **Chitin** (mostly homopolymer of NAG).

💡 **Bond Type:** Monosaccharides are linked together by **Glycosidic Bonds** formed by dehydration (loss of H_2O).

SECTION 9.6 NUCLEIC ACIDS

NCERT Page 111

Polynucleotide Structure

Nucleic acids are true biomacromolecules present in the acid-insoluble retentate. They are **polynucleotides** functioning as genetic material in living organisms.

3 Distinct Building Block Components:

1. Heterocyclic Nitrogenous Base
2. Monosaccharide (Pentose Sugar)
3. Phosphoric acid / Phosphate group

Bases & Pentose Sugars

Component	Varieties / Types
Purines	Substituted double-ring heterocyclic skeleton: Adenine (A) & Guanine (G)
Pyrimidines	Substituted single-ring skeleton: Cytosine (C) , Uracil (U) , Thymine (T)
Pentose Sugars	Ribose (in RNA) or 2'-deoxyribose (in DNA)

DNA vs RNA Comparison

DNA (Deoxyribonucleic Acid):

Sugar = 2'-deoxyribose. Bases = A, G, C, and **Thymine (T)**. Double-stranded secondary helix.

RNA (Ribonucleic Acid):

Sugar = Ribose. Bases = A, G, C, and **Uracil (U)**. Single stranded in most cases.

🔗 **Phosphodiester Bond** links adjacent nucleotides between 3'-OH of one sugar and 5'-phosphate of next sugar.

SECTION 9.7 STRUCTURE OF PROTEINS

NCERT Pages 111-112

1. PRIMARY STRUCTURE

Linear

The specific **positional sequence of amino acids** in a polypeptide chain.

- **Left End:** First amino acid = **N-terminal amino acid** (has free -NH_2).
- **Right End:** Last amino acid = **C-terminal amino acid** (has free -COOH).

2. SECONDARY STRUCTURE

Helical / Sheet

Local folding of the polypeptide chain into regular patterns due to hydrogen bonding.

- **α -Helix:** Polypeptide folded into a revolving staircase coil. In proteins, **only right-handed helices** are observed!

- **β-Pleated Sheet:** Adjacent chains aligned in parallel/antiparallel sheets.

3. TERTIARY STRUCTURE

3D Spherical

The long polypeptide chain is folded upon itself like a **hollow woolen ball** giving a 3-dimensional view.

⚡ **Mandatory Rule:** Tertiary structure is **absolutely necessary** for the biological activities (catalytic, binding) of proteins!

4. QUATERNARY STRUCTURE

Multi-subunit

Spatial architectural arrangement of multiple folded polypeptide subunits relative to one another.

Example — Human Haemoglobin (Hb):

Consists of **4 subunits**: Two identical α-subunits and two identical β-subunits (2α + 2β).

SECTION 9.8 ENZYMES

NCERT Pages 112-118

Enzyme Fundamentals & Active Site

- **Protein Nature:** Almost all enzymes are proteins. Exception: Catalytic nucleic acids are called **Ribozymes**.
- **Active Site:** Crevice or pocket in tertiary structure into which the substrate fits.
- **Inorganic vs Organic Catalysts:**
 - Inorganic catalysts work efficiently at high temperatures and pressures.
 - Enzymes get damaged at high temperatures (above 40°C).
 - **Thermophilic Exception:** Enzymes isolated from hot vents / sulfur springs retain catalytic activity at 80°C – 90°C.

9.8.1 Chemical Reactions & Rate Acceleration

Process Rate Definition:

$$\text{Rate} = \Delta P / \Delta t$$

Rate doubles or decreases by half for every 10°C change in either direction.

Carbonic Anhydrase Power:



- Uncatalysed: ~200 molecules/hour.
- Catalysed: ~600,000 molecules/second! (Speeded by **10 million times**).

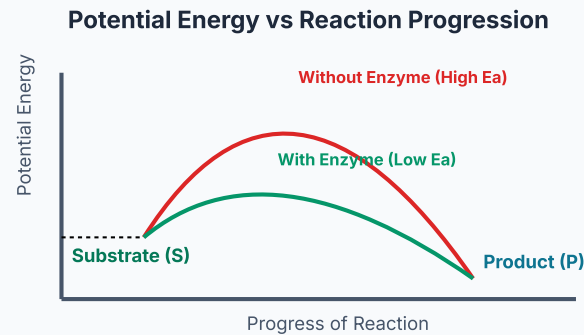
Glycolytic Pathway Variations (Glucose → 2 Pyruvic Acid):

Single pathway involving 10 enzyme-catalysed steps yields different end-products under varied conditions:

- **Aerobic conditions (human cells):** Pyruvic acid formed.
- **Anaerobic conditions (skeletal muscle):** Lactic acid formed.
- **Fermentation (yeast):** Ethanol (alcohol) formed.

9.8.2 Mechanism: Activation Energy & Transition State

Substrate (S) binds enzyme (E) at active site to form a transient **ES Complex**, passing through an unstable high-energy **Transition State** to yield Product (P).



- **Exothermic reaction:** P is at a lower energy level than S .
- **Activation Energy (E_a):** Difference between average energy content of S and transition state. Enzymes lower this energy barrier.

9.8.3 The 4-Step Catalytic Cycle of Enzyme Action

- STEP 1** Substrate binds to the active site of the enzyme, fitting precisely into the pocket.
- STEP 2** Binding of substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.
- STEP 3** Active site breaks substrate chemical bonds; new **Enzyme-Product (EP) complex** is formed.
- STEP 4** Enzyme releases free products; unaltered enzyme is ready to bind another substrate molecule.



9.8.4 Factors Affecting Enzyme Activity

1. Temperature & pH Optima

Enzymes display bell-shaped activity curves around an **Optimum Temperature** and **Optimum pH**.

- **Low Temperature:** Preserves enzyme in a temporarily inactive state.
- **High Temperature:** Destroys enzymatic activity due to heat denaturation of proteins.

2. Substrate Concentration $[S]$

Velocity rises initially until reaching maximum velocity (V_{max}). Further rise in $[S]$ does not increase rate because enzyme active sites become **fully saturated**.

K_m (**Michaelis Constant**): Substrate concentration at which velocity is $\frac{1}{2} V_{max}$.

3. Competitive Inhibition

NEET Must Know

Inhibitor closely resembles substrate in molecular structure and competes for the active site binding spot.

- **Classic Example:** Inhibition of **Succinic dehydrogenase** by **Malonate** (resembles substrate *succinate*).
- **Application:** Used in control of bacterial pathogens.

9.8.5 Classification and Nomenclature of Enzymes (6 Major Classes)

EC 4-Digit Classification System

1. Oxidoreductases / Dehydrogenases

Catalyse oxidation-reduction between two substrates.



2. Transferases

Catalyse transfer of a group (G, other than hydrogen) between substrates.



3. Hydrolases

4. Lyases

Catalyse cleavage/hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide, or P-N bonds using H₂O.

Catalyse removal of groups from substrates by mechanisms other than hydrolysis, leaving double bonds.



5. Isomerases

Catalyse inter-conversion of optical, geometric, or positional isomers.

6. Ligases

Catalyse linking together of 2 compounds (joining C-O, C-S, C-N, P-O bonds).

🔴 **MNEMONIC FOR 6 ENZYME CLASSES: "OTH LIL"** (Oxidoreductase, Transferase, Hydrolase, Lyase, Isomerase, Ligase)

9.8.6 Co-factors & Holoenzyme Complex

Enzymes can consist of protein polypeptide chains plus non-protein components called **co-factors** necessary for catalytic activity.

HOLOENZYME (Active Catalytic Complex) = APOENZYME (Protein Portion) + CO-FACTOR (Non-Protein Part)

Cofactor Type	Nature of Association	Key Examples & Vitamin Components
1. Prosthetic Groups	Organic compounds tightly bound to the apoenzyme.	Haem in <i>peroxidase</i> and <i>catalase</i> (catalyses $2H_2O_2 \rightarrow 2H_2O + O_2$). Haem forms part of the active site.
2. Co-enzymes	Organic compounds whose association with apoenzyme is transient (during catalysis). Serve as co-factors in multiple reactions.	Contain vitamins: NAD & NADP contain the vitamin Niacin .
3. Metal Ions	Form coordination bonds with side chains at active site and with substrate.	Zn²⁺ (Zinc) is cofactor for proteolytic enzyme Carboxypeptidase .

⚠️ **Crucial Fact:** When the co-factor is removed from the enzyme, catalytic activity is completely lost!

1. ACID-SOLUBLE POOL VS ACID-INSOLUBLE FRACTION

Feature	Acid-Soluble Pool (Filtrate)	Acid-Insoluble Retentate
Molecular Weight	18 to ~800 Daltons	>10,000 Daltons (except Lipids)
Molecules	Biomicromolecules (Amino acids, Monosaccharides, Nucleotides)	Biomacromolecules (Proteins, Nucleic Acids, Polysaccharides) + Lipids
Cellular Origin	Cytoplasmic composition	Macromolecules from cytoplasm & organelle membranes

2. PRIMARY VS SECONDARY METABOLITES

Feature	Primary Metabolites	Secondary Metabolites
Distribution	All living organisms (Animals, Plants)	Plants, Fungi, Microbes
Function	Direct, known physiological roles	Host physiological roles unclear; ecological/welfare use
Examples	Glucose, Ribose, Alanine, Adenine	Rubber, Morphine, Concanavalin A, Curcumin

FINAL REVISION & HIGH-YIELD MEMORY VAULT

NEET Quick Booster

⚡ Top 5 Abundance Facts

1. **Water (70-90%)**: Most abundant cellular compound.
2. **Protein (10-15%)**: Most abundant organic macromolecule.
3. **Collagen**: Most abundant protein in animal kingdom.
4. **RuBisCO**: Most abundant protein in entire biosphere.
5. **Cellulose**: Most abundant organic polymer on Earth.

💡 Bond Type Quick Summary

- **Peptide Bond:** Between -COOH and -NH_2 of amino acids (Protein).
- **Glycosidic Bond:** Between monosaccharides (Polysaccharides).
- **Phosphodiester Bond:** Between 3'-OH and 5'-phosphate of nucleotides (DNA/RNA).
- **Ester Bond:** Between glycerol hydroxy groups and fatty acid carboxyl groups (Triglyceride).

⚠ **Key Exception Checklist**

- **Lipids:** MW <800 Da, but insoluble due to membrane vesicles.
- **Ribozyme:** Non-protein enzyme (RNA catalyst).
- **Thermophiles:** Enzymes active up to 80°C – 90°C .
- **Right-handed Helix:** Proteins only have right-handed helices.



NCERT TEXTBOOK EXERCISES (SOLVED SUMMARY)

11 Questions Covered

Q1: What are macromolecules? Give examples.

Biomolecules found in acid-insoluble retentate with molecular weights above 10,000 Daltons. Examples: Proteins, Nucleic Acids, Polysaccharides.

Q2: What is meant by tertiary structure of proteins?

3-dimensional folding of polypeptide chain upon itself like a hollow woolen ball. Essential for biological activity of enzymes.

Q5: Explain the composition of triglyceride.

Composed of one molecule of glycerol (trihydroxy propane) esterified with three fatty acid chains (R_1 , R_2 , $\text{R}_3\text{-COOH}$).

Q7: Draw/Describe the structure of alanine.

α -amino acid where variable R group is a methyl group (-CH_3). Formula: $\text{H}_2\text{N-CH(CH}_3\text{)-COOH}$.

Q8: What are gums made of? Is Fevicol different?

Natural gums are secondary metabolite polysaccharides. Fevicol is a synthetic polymer (polyvinyl acetate adhesive), not a polysaccharide.

Q11: Describe important properties of enzymes.

Proteinaceous nature, substrate specificity, lower activation energy, sensitive to optimum Temp & pH, presence of active sites, subject to inhibition.